Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-15: 1 Class

Please listen Carefully
&
Keep your phone(s) on silent mode

Why Engineer Cell Signaling

- seem like an audacious and foolish goal
- engineering of cell signaling is not simply a process for applying an already well-developed understanding
- ‘understanding by building’
- Engineering new cell signaling networks offers an approach for us to test and expand our understanding of the organizational principles of complex molecular systems

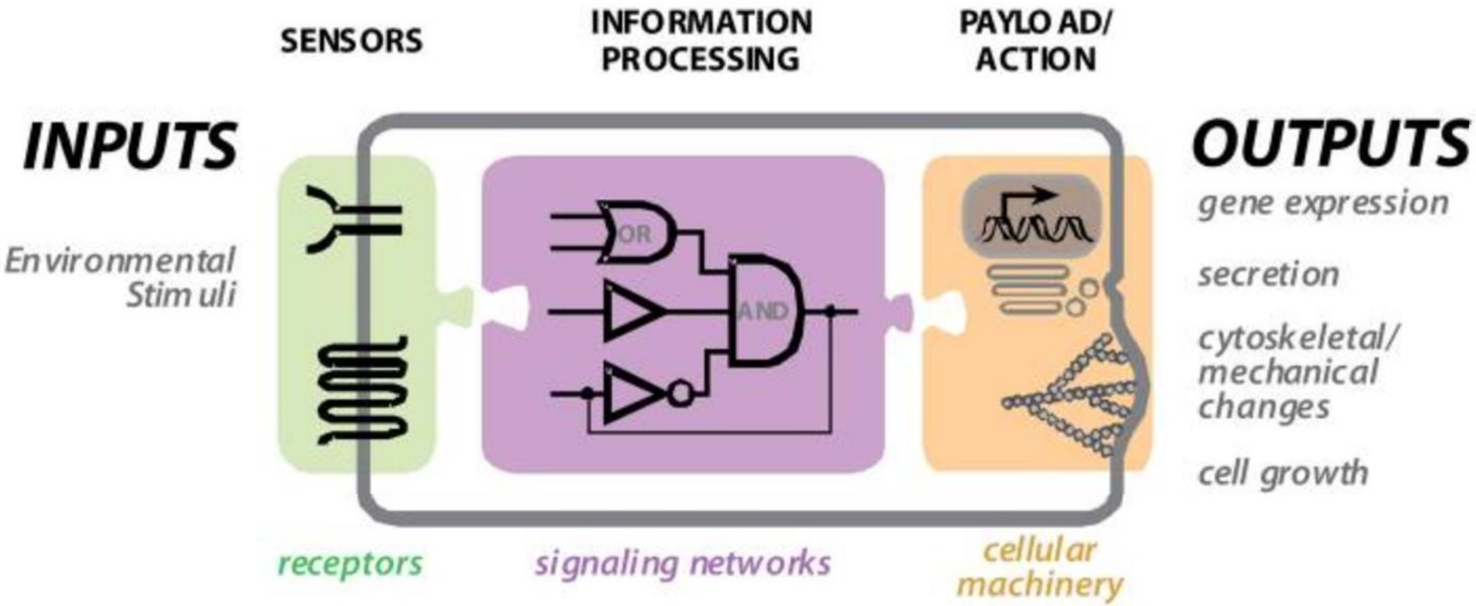
Applications of engineered signaling in therapy and biotechnology

- **Designed anti-cancer cells**
  - ❑ Immune cells such as T-lymphocytes or Natural Killer cells could be modified to identify and kill tumor cells
  - ❑ Designed cells that detect these tumor specific inputs could be engineered to yield a number of different responses,
    - imaging agents that aid in identifying tumors and metastases, and
    - the control of endogneous immune cell responses, such as chemotaxis,
      phagocytosis and cell killing.
    - programmed to secrete factors that disrupt the local tumor microenvironment, such as pro inflammatory cytokines and anti-angiogenesis factors, making it untenable for sustained tumor growth.
- **Targeted immunosupression**
  - An immune cell could also be designed to block

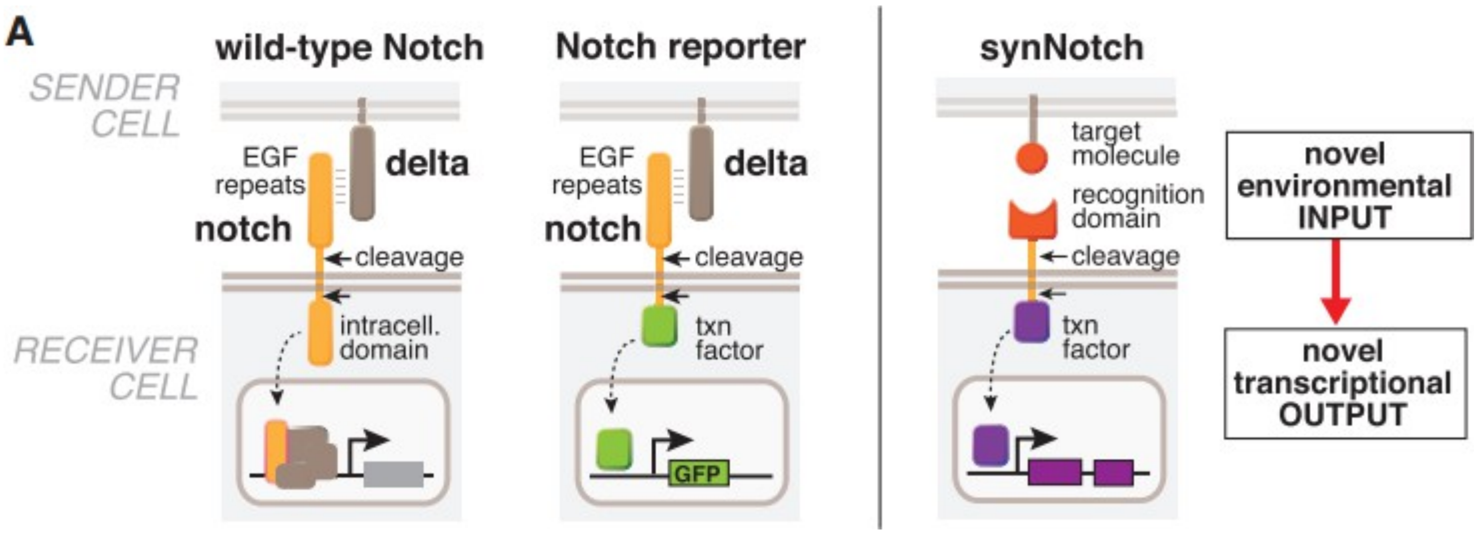
Are Cell Signaling Networks Engineerable?

- A hallmark of signaling proteins, which is thought to play a major role in evolution, is their modular structure.
- if we could understand how evolution works with these modules, we might be able to exploit the same toolkit to find regions of behavior space that evolution has, to our knowledge, not yet explored.

Engineering New Sensor System

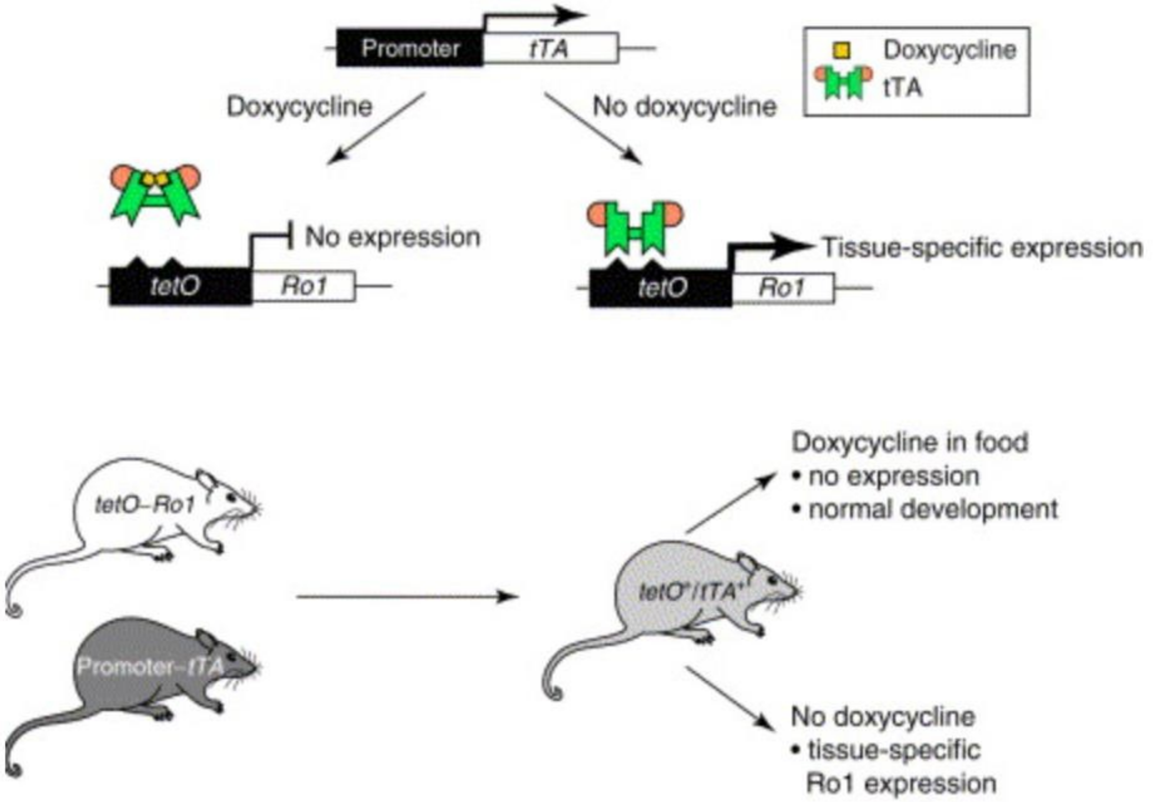


Redirecting the output of natural receptor



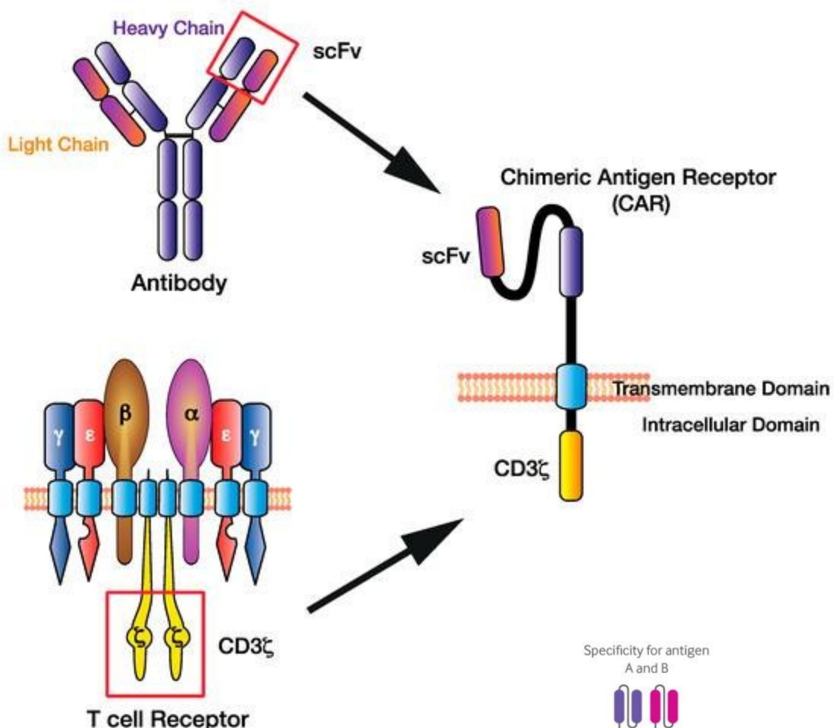
Receptors that detect novel small molecule input

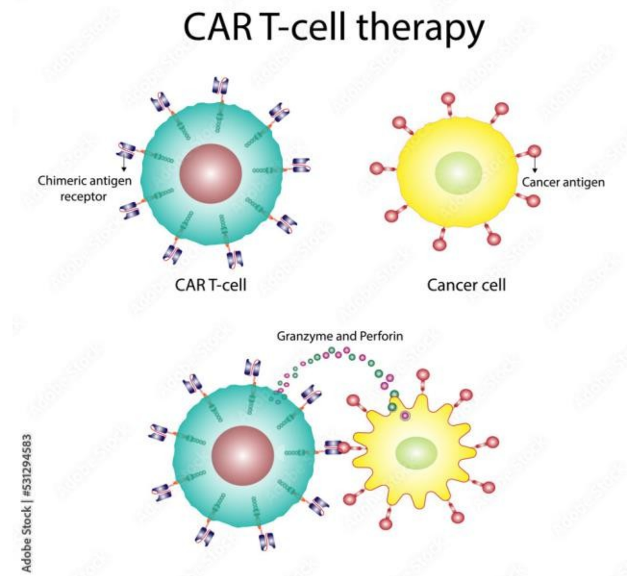
Receptors activated solely by synthetic ligands (RASSLs)

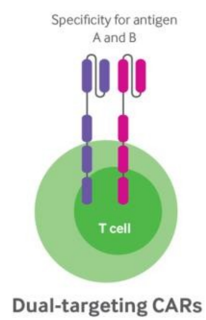


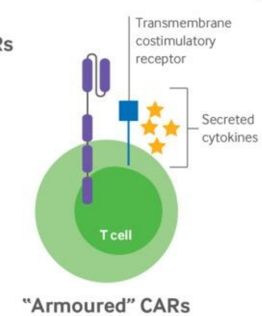
TRENDS in Pharmacological Sciences

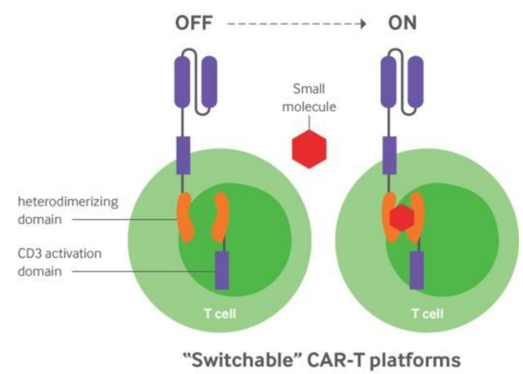
Receptors that detect user specified antigens



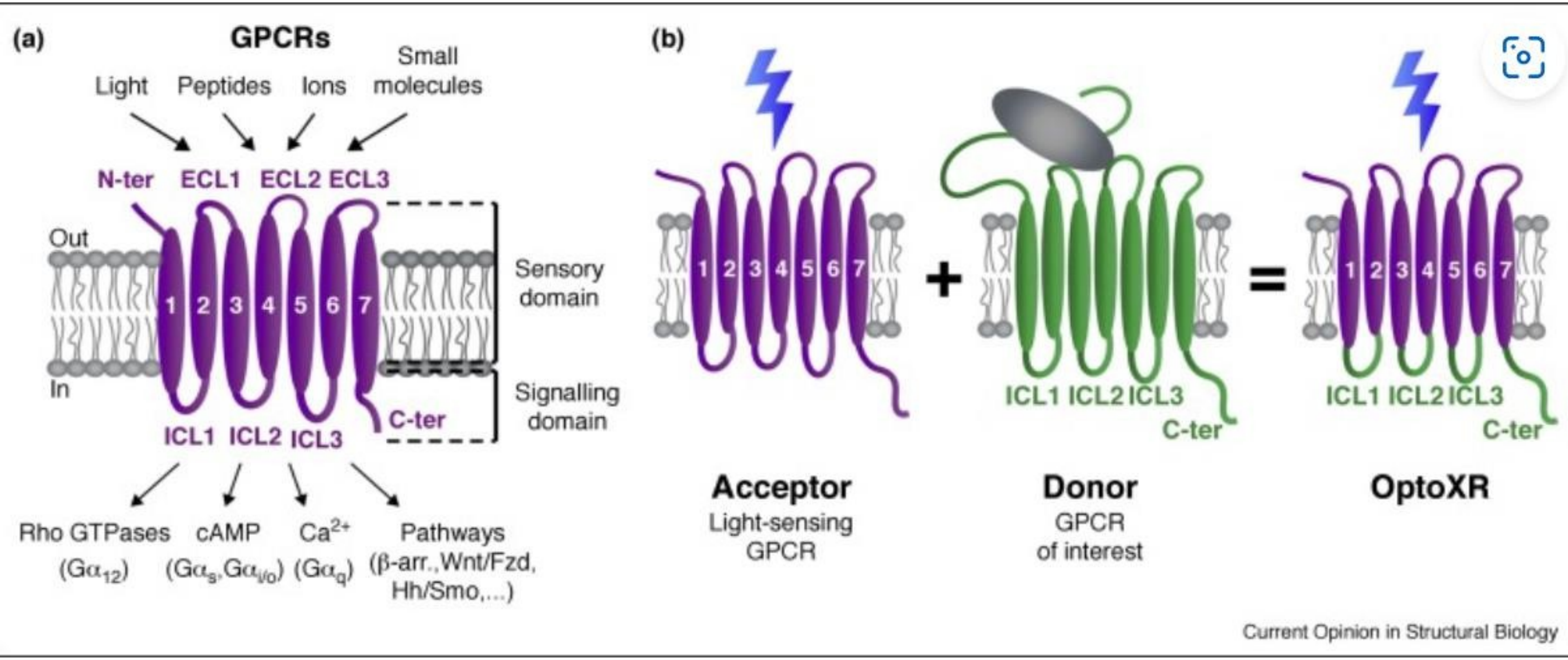








Sensors that detect physical signals such as light

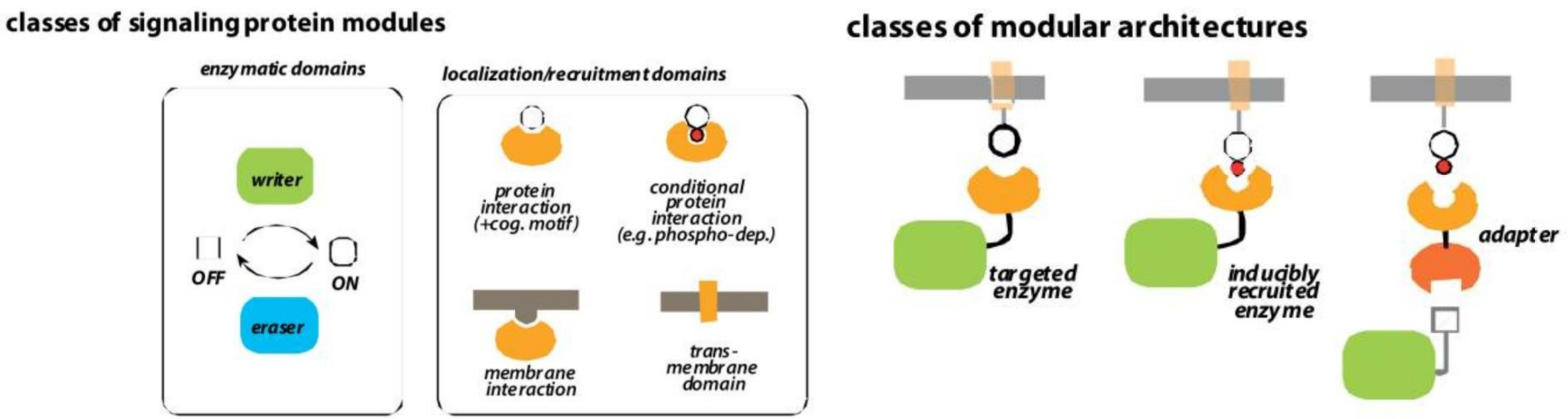


Engineering Signal Processing Systems

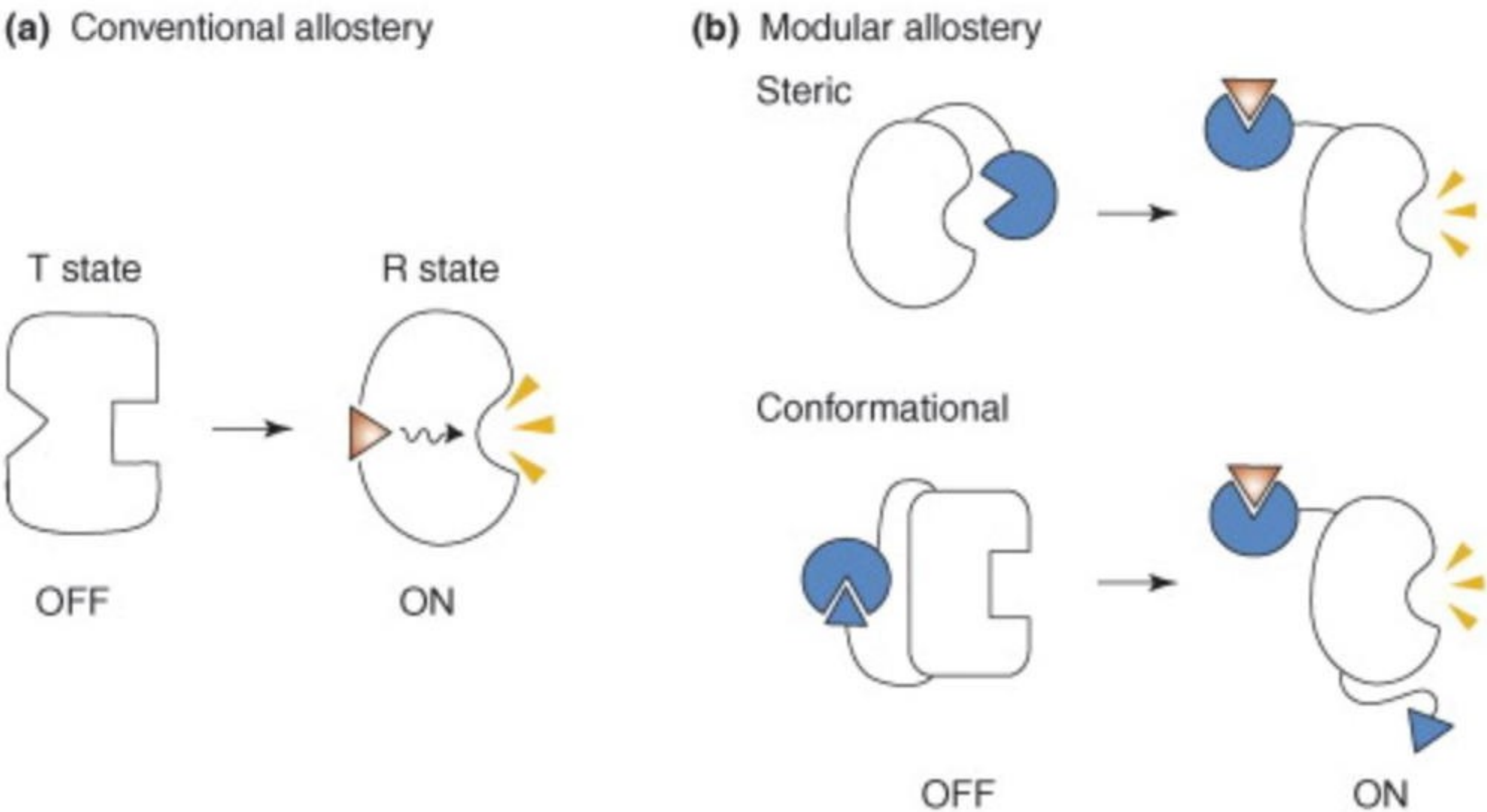
- Modular logic of signal processing

Intracellular signaling proteins are highly modular.
Most modules fall into two classes:

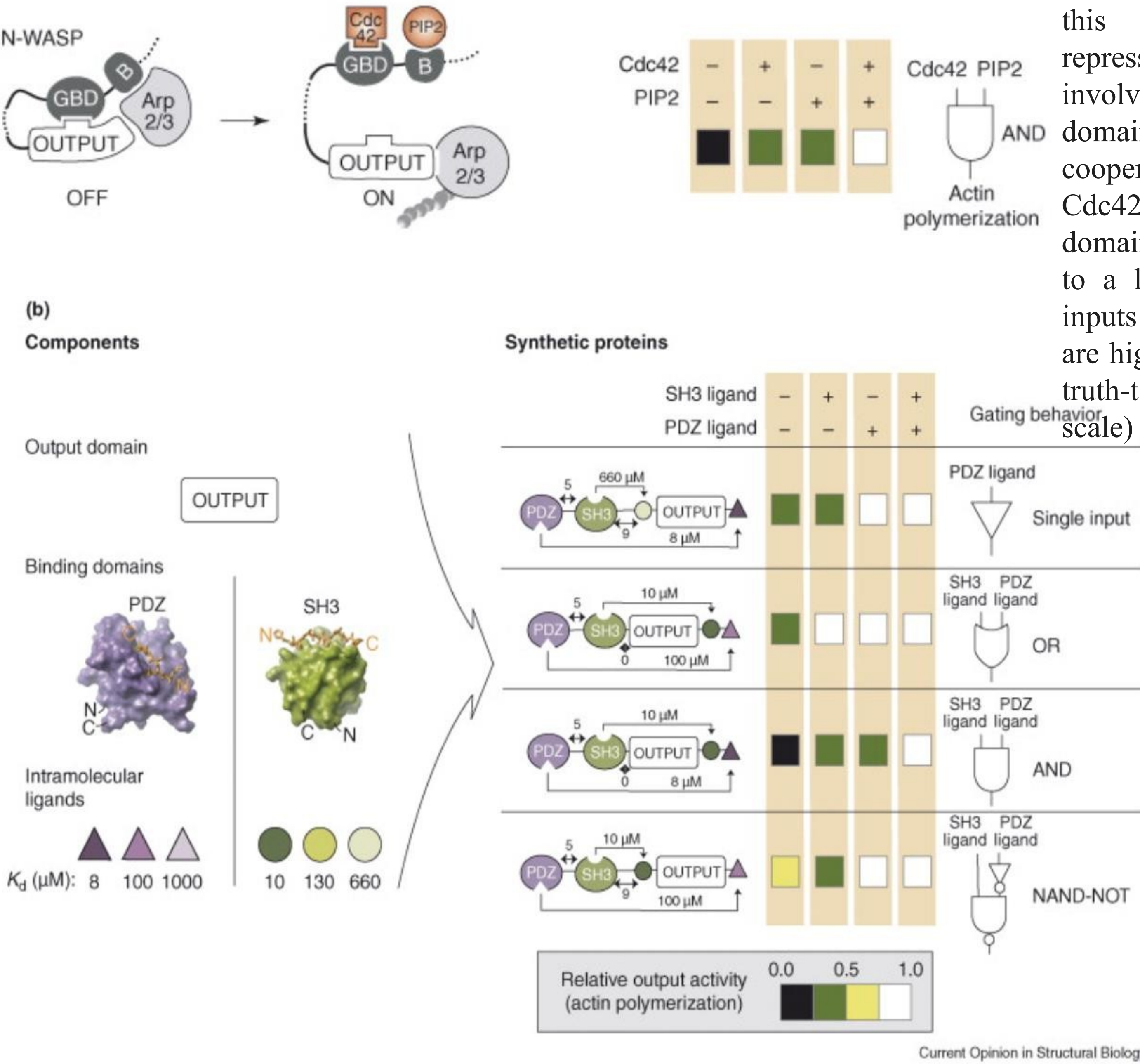
1. Enzymatic domains, such as kinases and phosphatases, which catalyze the post-translational modifications or
2. conformational changes by which information is stored.



Engineering new protein switches



Current Opinion in Structural Biology

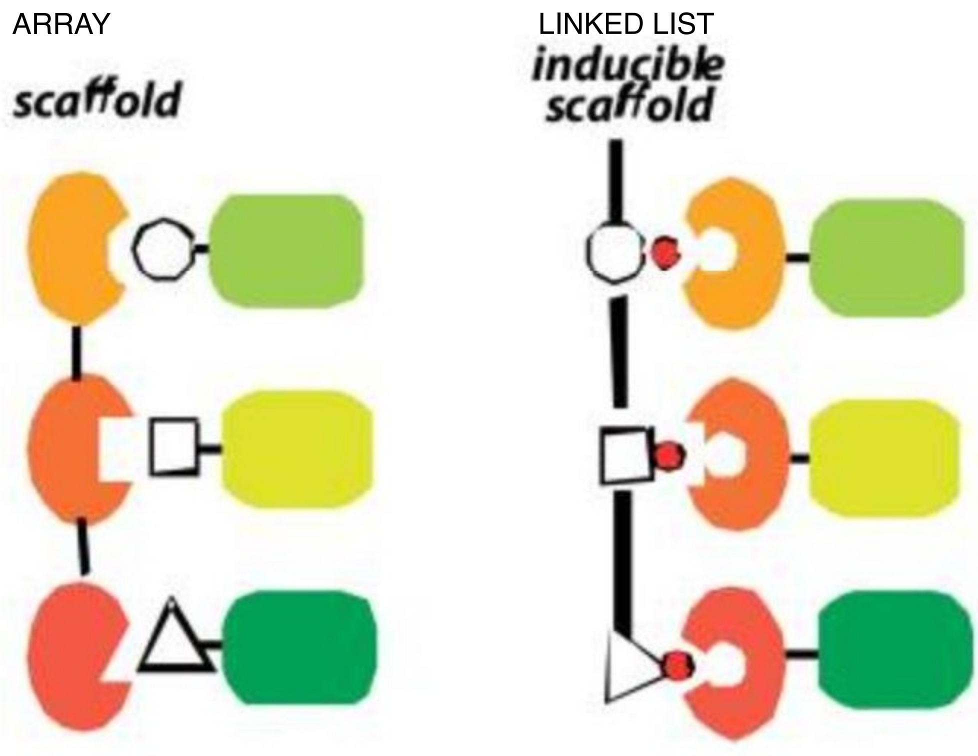


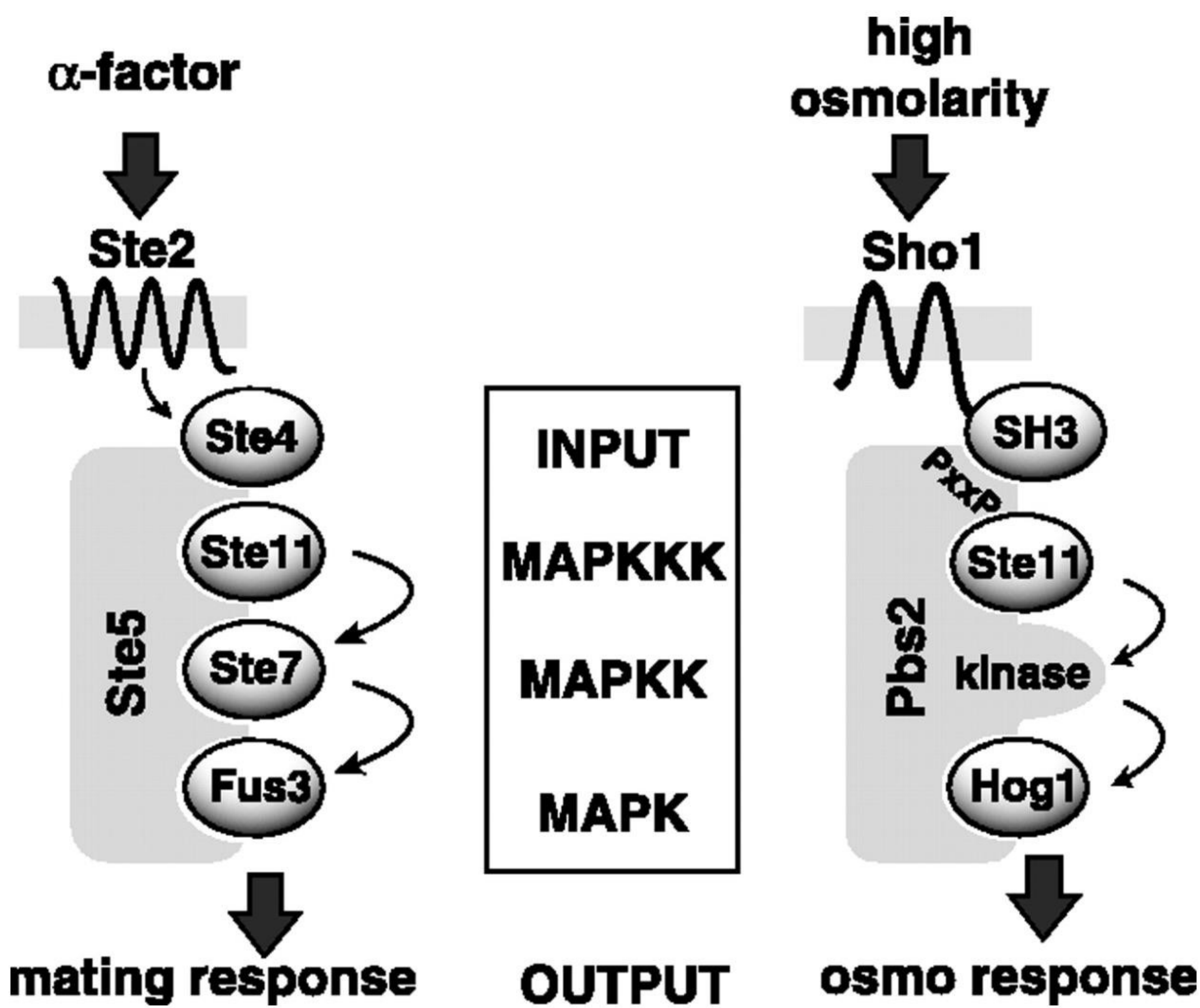
The output domain of N-WASP can activate actin polymerization via the Arp2/3 complex. Under basal conditions, this domain is conformationally repressed by autoinhibitory interactions involving the GBD and basic (B) domain. These interactions are cooperatively relieved by the binding of Cdc42 and PIP2 to these respective domains. Thus, N-WASP behaves akin to a logical AND gate: the individual inputs are poor activators, but together are highly potent (activity is indicated in truth-table format using color-coded scale)

Dueber *et al*. attempted to reprogram the N-WASP output domain by imposing synthetic autoinhibitory interactions using SH3 and PDZ domains. A library of potential gates was constructed using different domain architectures, and intramolecular SH3 and PDZ domain ligands of varying affinity. The right panel shows examples of diverse gating behaviors generated

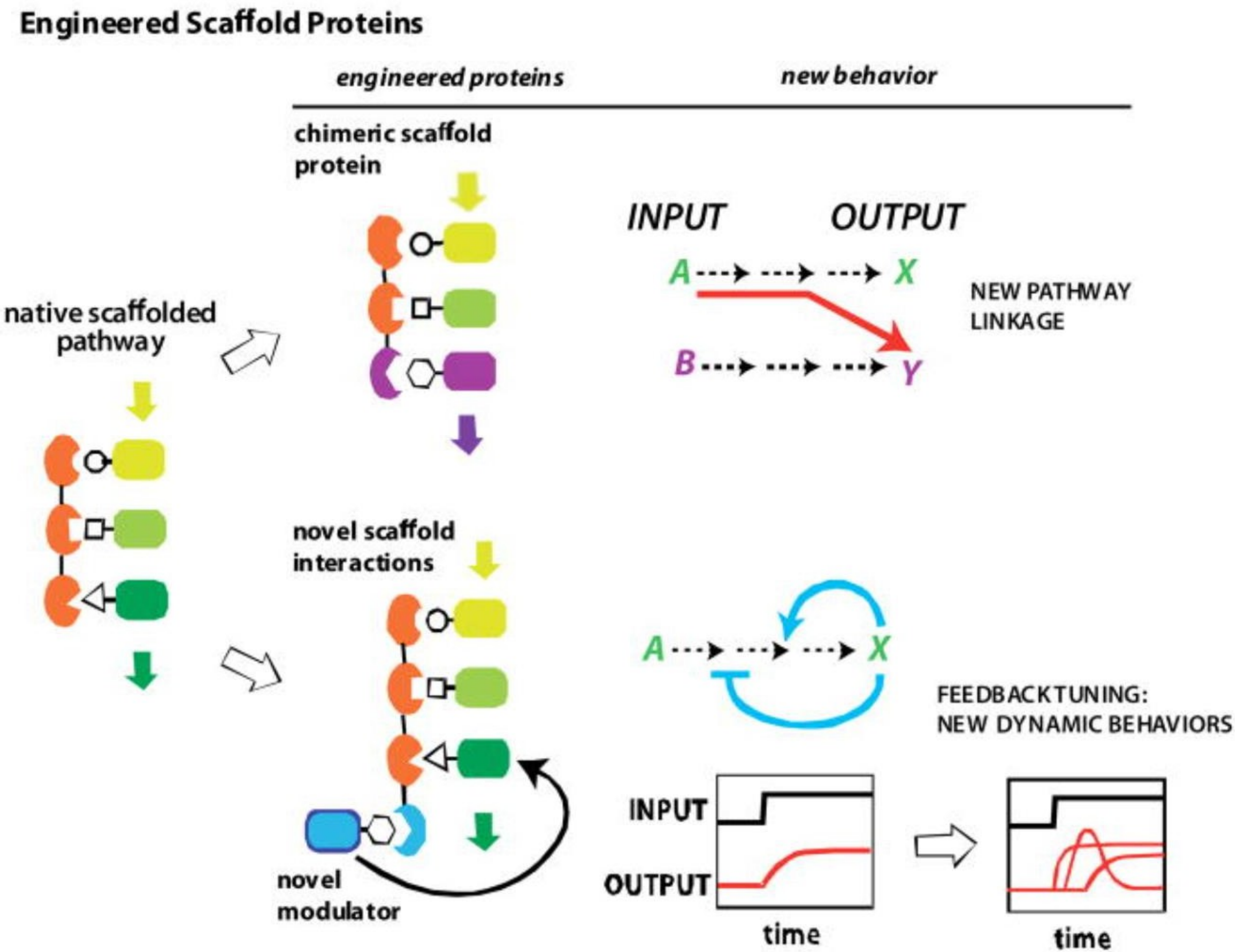
Scaffold proteins as molecular circuitboards

Scaffold proteins have been shown to not only mediate the linear input/output relationship of pathways, but also to coordinate the recruitment of modulatory factors that shape the dose dependence and dynamics of pathway response

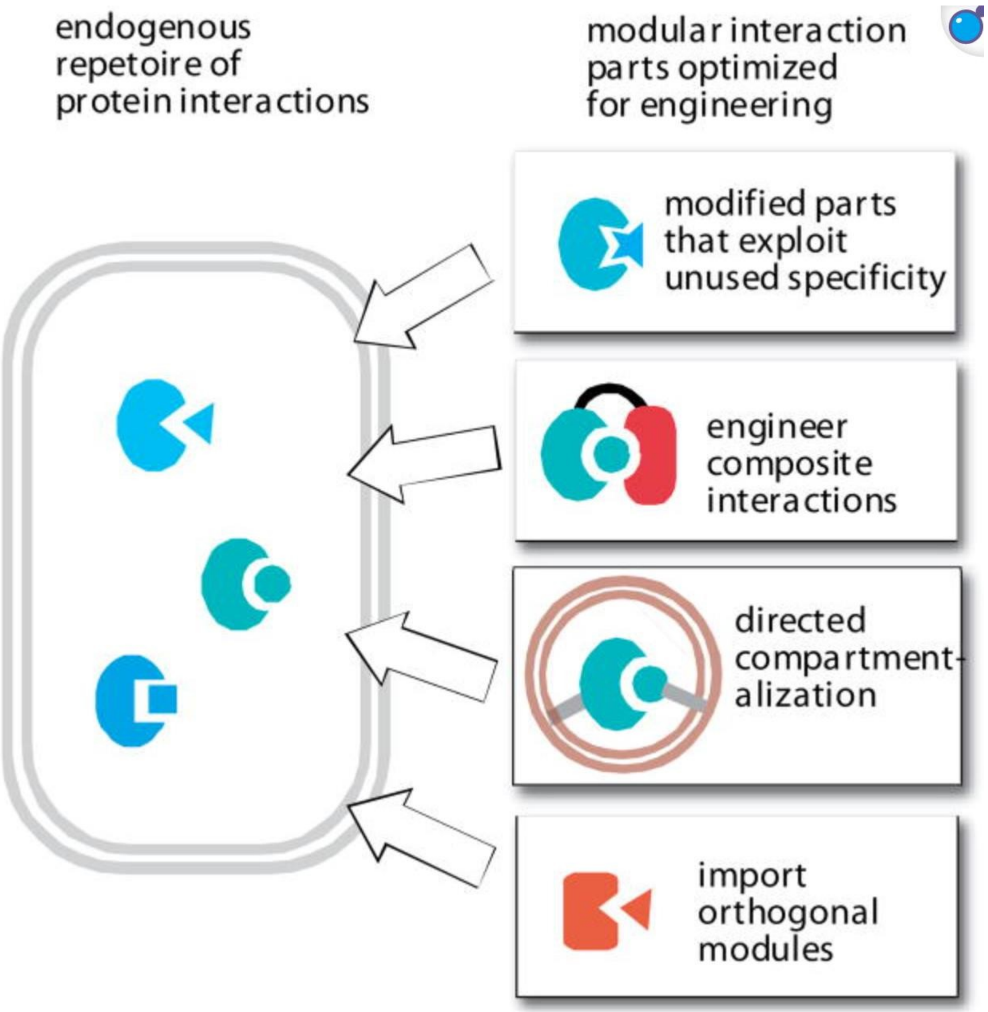




α-factor
Ste2
Ste4
Ste11
Ste7
Fus3
Ste5
mating response
INPUT
MAPKKK
MAPKK
MAPK
OUTPUT
high
osmolarity
Sho1
SH3
PxxP
Ste11
Pbs2
kinase
Hog1
osmo response



Engineered Scaffold Proteins
engineered proteins
new behavior
chimeric scaffold protein
INPUT
OUTPUT
A
X
B
Y
NEW PATHWAY LINKAGE
native scaffolded pathway
novel scaffold interactions
A
X
FEEDBACK TUNING: NEW DYNAMIC BEHAVIORS
INPUT
OUTPUT
time
time
novel modulator



A native cell has its own native repetoire of protein interaction modules, and thus it is challenging to engineer new functions using related interaction modules that might show inadvertent and unintended crosstalk in the cell. An optimized toolkit of interaction parts could significantly increase the predictability of cellular engineering, by eliminating the chance for unintended crosstalk. Several strategies for optimization include engineering of interaction modules that exploit untapped specificity; engineering of composite, multi-domain interactions; combining interaction modules with novel subcellular targeting; and importing orthogonal interaction modules (either synthetically